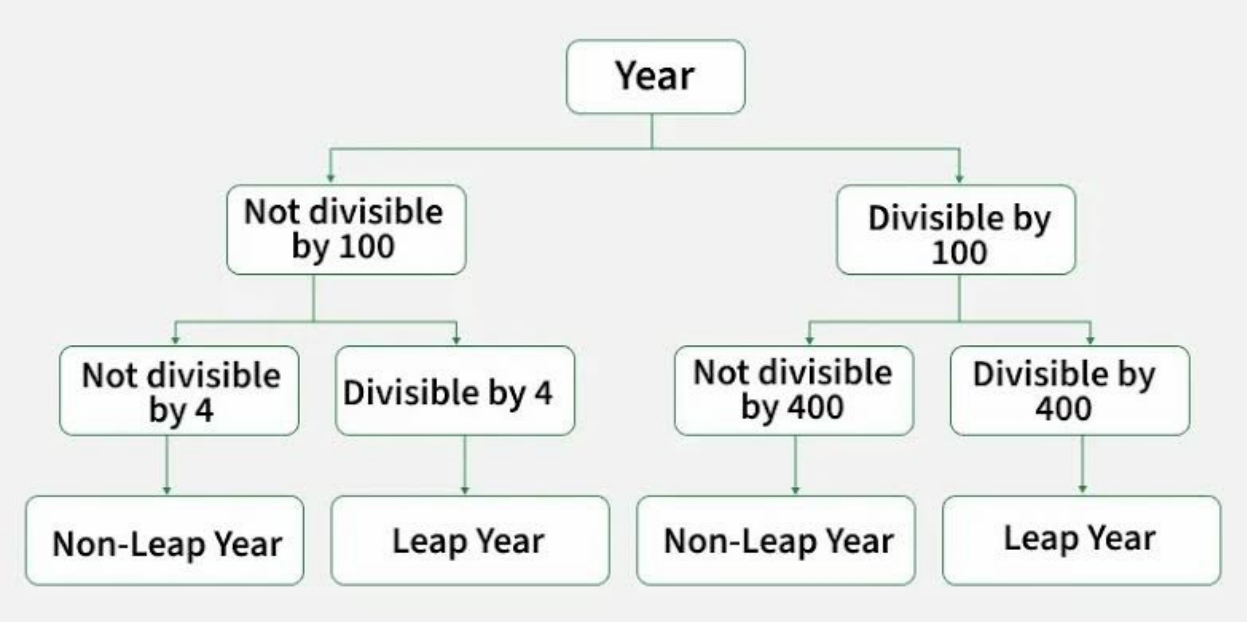
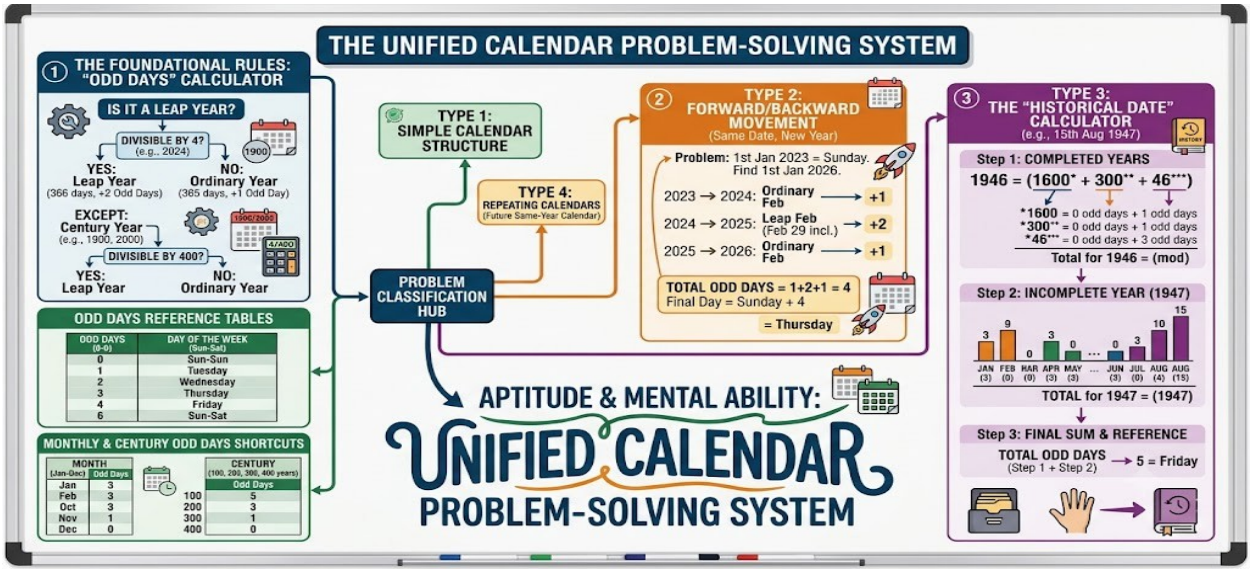
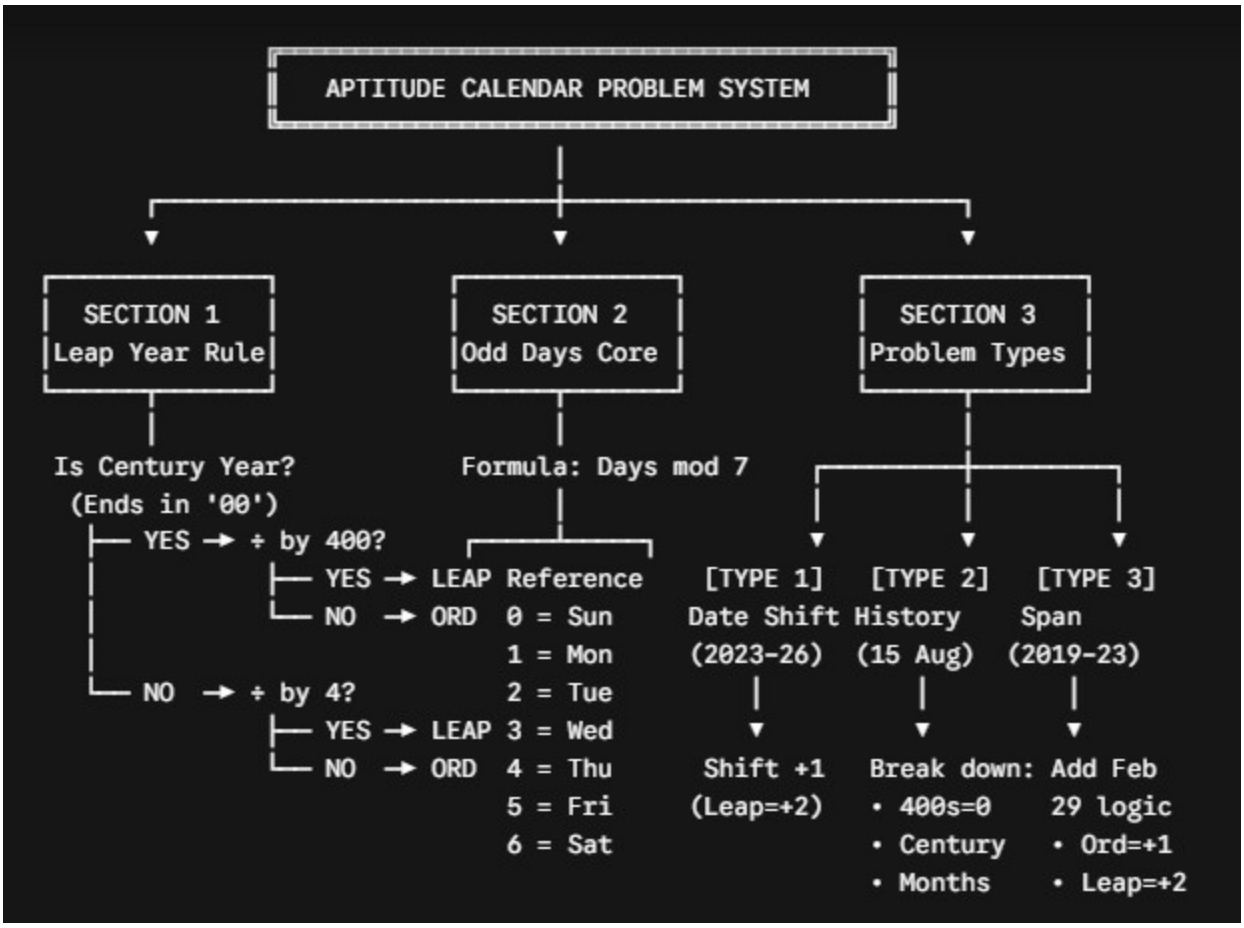
Aptitude & Mental Ability Lecture Notes: Calendar Problems







1. Structure of the Gregorian Calendar

- **Ordinary Year:** Consists of **365 days** (52 weeks +1 day).
- **Leap Year:** Consists of **366 days(52 weeks +2 days**) due to an extra day in February (29 days).

2. Century vs. Non-Century Leap Year Rules

- **Non-Century Years:** A non-century year is a leap year if it is divisible by **4** (e.g., 2024, 2028, 2032).
- **Century Years:** A century year (ending in `00`) is a leap year **only if it is divisible by 400**.
- A **Century Year** is any year that marks the end of a 100-year cycle in the calendar.

Examples of Century Years

- **Past Century Years:**1600, 1700, 1800, 1900, 2000
- **Future Century Years:**2100, 22002300, 2400

🔔Why 1900 wasn't a leap year, but 2000 was:

- 1900 /400 = 4.75 (Not divisible by 400 =**Ordinary Year**)
- 2000 / 400 = 5 (Divisible by 400 =**Leap Year**)
- Similarly, 1700, 1800, and 1900 were ordinary years, whereas 1600 and 2000 were leap years.

3. Odd Days Concept

An **Odd Day** is the remainder left when the total number of days is divided by 7 (a week cycle).

- Odd days = Total days mod 7
-
- **1 Ordinary Year** = 365 mod 7 = 1 odd day
- **1 Leap Year** = 366 mod 7 = 2 odd days

1. Month-wise Odd Days (Days mod 7)

Month	Days	Odd Days	Shortcut Note
Jan, Mar, May, Jul, Aug, Oct, Dec	31	**3**	31 mod 7 = 3
Apr, Jun, Sep, Nov	30	**2**	30 mod 7 = 2
Feb (Ordinary Year)	28	**0**	28 mod 7 = 0
Feb (Leap Year)	29	**1**	29 mod 7 = 1

2. Century-wise Odd Days Shortcut

Year Interval	Calculation Breakdown	Final Odd Days
100 Years	76 Ordinary + 24 Leap = 76(1) + 24(2) = 124 mod 7	**5**
200 Years	5 times 2 = 10 mod 7	**3**
300 Years	5 times 3 = 15 mod 7	**1**
400 Years	5 * 4 + 1 (extra leap year) = 21\mod 7	**0**

Standard Odd Days Reference Table:

Total Odd Days	Day of the Week
0	Sunday
1	Monday
2	Tuesday
3	Wednesday
4	Thursday
5	Friday
6	Saturday

Condition	Odd Days Added	Example
Normal Year Shift	$+1$ Day	1 Jan 2022 (Sat) $\to$ 1 Jan 2023 (Sun)
Leap Year Shift*(Feb 29 crossed)*	$+2$ Days	1 Jan 2024 (Mon) $\to$ 1 Jan 2025 (Wed)
Century Year Divisible by 400 ($1600, 2000, 2400$)	Leap Year ($+2$)	Included in standard leap count
Century Year NOT Divisible by 400 ($1700, 1800, 1900, 2100$)	Non-Leap Year ($+1$)	Treated as a normal year

4. Forward Movement of Days

When moving forward from a given date in year Y to the same date in year Y+1:

- Shift forward by +1 day if the intervening period contains an ordinary February (28 days).
- Shift forward by +2 days if the intervening period contains a leap February (Feb 29).

Example:

- 1st Jan 2023 = **Sunday**
- 1st Jan 2024 = **Monday** (+1 day, as 2023 is an ordinary year)
- 1st Jan 2025 = **Wednesday** (+2 days, as 2024 is a leap year and includes Feb 29)

Question 1: Year-Shift Logic

If 1st January 2023 was a Sunday, what day of the week was 1st January 2026?

Solution:

1. **1st Jan 2023 to 1st Jan 2024:** 2023 is an ordinary year = +1 odd day.
2. **1st Jan 2024 to 1stJan 2025:** 2024 is a leap year (includes Feb 29, 2024) = 2 odd days.
3. **1st Jan 2025 to 1st Jan 2026:** 2025 is an ordinary year = +1 odd day

Total Odd Days = 1 + 2 + 1 = 4 days

Day on 1st Jan 2026 = Sunday +4 day = Thursday

Problem 4: Multi-Decade Shift (LRDI Aptitude Standard)

Question: If 26th January 2000 was a Wednesday, what day of the week was 26th January 2020?

If 29 April 1995 was a Saturday , what day of the week was 29 April 2023 ?

Solution:

1. Total years between 2000 and 2020: Total Years = 2020 - 2000 = 20 years
2. Identify Leap Years: Potential leap years in this range: 2000, 2004, 2008, 2012, 2016.

- Check 2000: The period starts at 26th Jan 2000, so 29th Feb 2000 is included. Therefore, 2000 counts as a leap year.
- Check 2020: The period ends at 26th Jan 2020, so 29th Feb 2020 is NOT included. Therefore, 2020 is not counted in this interval.

Total Leap Years = 5 (2000, 2004, 2008, 2012, 2016) Total Non-Leap Years = 20 - 5 = 15

3. Calculate total odd days: Each non-leap year gives 1 odd day. Each leap year gives 2 odd days.

Total Odd Days = (15 non-leap years * 1) + (5 leap years * 2) Total Odd Days = 15 + 10 = 25 odd days

4. Simplify modulo 7: 25 / 7 = 3 weeks with a remainder of 4 odd days. 25 mod 7 = 4 odd days.
5. Shift the day forward: Wednesday + 4 days = Sunday (Wednesday -> Thursday [1] -> Friday [2] -> Saturday [3] -> Sunday [4])

Answer: Sunday

Question 2: Finding the Day for an Exact Historical Date

What day of the week was 15st August 1947? // 21st July 1961 ?

Solution:

Step 1: Break down the completed years up to 1946

- **1946 years** = 1600 years + 300 years + 46 years
- **1600 years:** 0 odd days *(Multiple of 400)*
- **300 years:** 1 odd day
- **46 years:**
  - Leap years = 46 ÷ 4 = 11
  - Ordinary years = 46 - 11 = 35
  - Odd days = (11 × 2) + (35 × 1) = 22 + 35 = 57 days
  - 57 mod 7 = 1 odd day *(57 ÷ 7 leaves a remainder of 1)*

Odd days up to 1946 = 0 + 1 + 1 = **2 odd days**

Step 2: Calculate odd days in 1947 up to 15th August

- Jan (31 days) = 3 odd days
- Feb (28 days) = 0 odd days
- Mar (31 days) = 3 odd days
- Apr (30 days) = 2 odd days
- May (31 days) = 3 odd days
- Jun (30 days) = 2 odd days
- Jul (31 days) = 3 odd days
- Aug (15 days) = 1 odd day

Total days in 1947 = 3 + 0 + 3 + 2 + 3 + 2 + 3 + 1 = 17 days

17 mod 7 = **3 odd days***(17 ÷ 7 leaves a remainder of 3)*

Step 3: Sum all odd daysTotal Odd Days = 2 + 3 = **5 odd days**

Referring to the odd days reference table, **5 = Friday**.

Answer: 15th August 1947 was a **Friday**.

Question 3: Same-Date Calculation Across Multiple Years

If 15th March 2019 was a Friday, what day of the week was 15th March 2023?

Solve : If 15th April 2012 was Sunday, what day of the week was 15th April 2025 ?

Solution:

Count the odd days year by year from March 15, 2019 to March 15, 2023:

- **2019 to 2020:** February 2020 (29 days) is included ->**2 odd days**
- **2020 to 2021:** February 2021 (28 days) ->**1 odd day**
- **2021 to 2022:** February 2022 (28 days) ->**1 odd day**
- **2022 to 2023:** February 2023 (28 days) ->**1 odd day**

Total Odd Days = 2 + 1 + 1 + 1 = **5 days**

Day = Friday + 5 days = **Wednesday**

Question: If 15th August 2017 was a Tuesday, what day of the week was 15th August 2018?

Solution:

Understand the time gap: From 15th August 2017 to 15th August 2018 is exactly 1 full year (365 days).

Check for February 29th: The February that occurs between August 2017 and August 2018 belongs to the year 2018. Since 2018 is an ordinary year, February has 28 days.

Calculate Odd Days: 1 Ordinary Year = 1 Odd Day.

Shift the Day: Add 1 day to the original day (Tuesday).

Tuesday + 1 day = Wednesday.

Final Answer: 15th August 2018 was a Wednesday.

Example 3: Day Shift Involving a Leap Year

Question: If 1st January 2024 was a Monday, what day of the week was 1st January 2025?

Detailed Step-by-Step Solution:

Understand the time gap: From 1st January 2024 to 1st January 2025 is exactly 1 year.

Check for February 29th: During this period, we cross February 2024. Since 2024 is divisible by 4 (2024 / 4 = 506), it is a leap year with 29 days in February.

Calculate Odd Days: A leap year contains 366 days (52 weeks + 2 extra days). Therefore, it contributes 2 Odd Days.

Shift the Day: Add 2 days to the original day (Monday).

Monday + 2 days = Wednesday.

Final Answer: 1st January 2025 was a Wednesda

Example 4: Multi-Year Day Calculation

Question: If 8th March 2010 was a Monday, what day of the week was 8th March 2015?

If 2nd Aug 1945 was a Thursday , what day of week was 2th Aug 2011 ?

Detailed Step-by-Step Solution:

Find the total number of years: Total Years = 2015 - 2010 = 5 years.

Count Ordinary Years and Leap Years in this span: 2012 is a leap year (1 leap year). 2010, 2011, 2013, 2014 are ordinary years (4 ordinary years).

Calculate total odd days contributed: 4 Ordinary Years = 4 x 1 = 4 odd days. 1 Leap Year = 1 x 2 = 2 odd days. Total Odd Days = 4 + 2 = 6 odd days.

Shift the Day: Add 6 days to Monday.

Monday + 6 days = Sunday.

Final Answer: 8th March 2015 was a Sunday.

3. Practice Assignment Questions with Elaborated Solutions

Section A: Short Conceptual Questions

Question 1

Question: Why is 1900 not considered a leap year, whereas 2000 is a leap year? Explain clearly.

Detailed Answer:

The solar year (earth's rotation around the sun) takes approximately 365.2422 days, not exactly 365.25 days. Adding a leap year every 4 years adds slightly too much time over centuries.

To correct this error, calendar rules state that a centennial year (a year ending in '00') is counted as a leap year only if it is divisible by 400.

For 1900: 1900 / 400 = 4.75. Since it is not evenly divisible by 400, it was treated as an ordinary year with 365 days.

For 2000: 2000 / 400 = 5. Since it is completely divisible by 400, it was a leap year with 366 days.

Question 2

Question: How many weeks and odd days are there altogether in a block of 3 consecutive ordinary (non-leap) years?

Detailed Solution:

Calculate total days: 1 Ordinary Year = 365 days. 3 Ordinary Years = 365 x 3 = 1095 days.

Convert total days into weeks and odd days: Divide 1095 by 7: 1095 / 7 = 156 weeks with a remainder of 3 days.

Alternative Odd Day method: Each ordinary year has 1 odd day. Total odd days = 1 + 1 + 1 = 3 odd days.

Final Answer: 156 complete weeks and 3 odd days.

Question 3

Question: A non-leap year begins on a Friday (1st January). On which day of the week will it end (31st December)? Explain why.

Detailed Solution:

An ordinary year has 365 days. 365 days = 52 complete weeks + 1 odd day.

If Day 1 (1st January) is Friday: The 1st week ends on Thursday (Day 7). The 52nd week ends on Thursday (Day 364).

The remaining 365th day (31st December) becomes the extra day, which is Friday.

Key Concept to Remember: Any non-leap year always starts and ends on the exact same day of the week.

Section B: Application Problems

Question 4

Question: If 26th January 2020 was a Sunday, what day of the week was 26th January 2021?

Detailed Solution:

Identify the interval: From 26th January 2020 to 26th January 2021 is 1 year.

Determine if Feb 29 is included: The year 2020 is a leap year (2020 / 4 = 505). Since our date range goes from Jan 2020 to Jan 2021, it includes 29th February 2020.

Determine Odd Days: Because 29th Feb 2020 is included, this 1-year period contributes 2 Odd Days.

Shift the Day: Add 2 days to Sunday.

Sunday + 2 days = Tuesday.

Final Answer: 26th January 2021 was a Tuesday.

Question 5

Question: If 10th July 2013 was a Wednesday, what day of the week was 10th July 2019?

Detailed Solution:

Calculate total span in years: 2019 - 2013 = 6 years.

Identify leap years in this span: The years between July 2013 and July 2019 are 2014, 2015, 2016, 2017, 2018, 2019. 2016 is a leap year (2016 / 4 = 504). Therefore, there is 1 Leap Year (2016) and 5 Ordinary Years.

Calculate total odd days: 5 Ordinary Years = 5 x 1 = 5 odd days. 1 Leap Year = 1 x 2 = 2 odd days. Total Odd Days = 5 + 2 = 7 odd days.

Simplify odd days: Divide 7 by 7 to get the remainder: 7 / 7 = 0 remainder (0 Odd Days).

Shift the Day: Add 0 days to Wednesday.

Wednesday + 0 days = Wednesday.

Final Answer: 10th July 2019 was a Wednesday.

Question 6

Question: Will the calendar for the year 2023 be completely identical to the calendar for the year 2034? Prove your answer with complete odd day calculations.

Detailed Solution:

Two years have identical calendars if:

1. They are both the same type of year (both ordinary OR both leap years).
2. The total number of odd days between them is a multiple of 7 (i.e., 0 remainder).

Step 1: Check Year Types 2023 is an ordinary year. 2034 is an ordinary year (2034 / 4 = 508.5). Both match.

Step 2: Calculate Years and Odd Days Total span = 2034 - 2023 = 11 years. Identify Leap Years between 2023 and 2034: 2024, 2028, 2032 (3 leap years). Ordinary Years: 11 - 3 = 8 ordinary years.

Step 3: Calculate Total Odd Days 8 Ordinary Years = 8 x 1 = 8 odd days. 3 Leap Years = 3 x 2 = 6 odd days. Total Odd Days = 8 + 6 = 14 odd days.

Step 4: Check Remainder Divide 14 by 7: 14 / 7 = 2 weeks with 0 remainder.